Supplementary Methods: Constrained Bivariate Gaussian Mixture Modeling

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1. Objective and Input Data

To determine whether the mRNA–protein response regimes identified by Differential Pattern Analysis (DiPa) reflect distinct biological subpopulations rather than arbitrary slices of a continuous cloud, we fitted constrained bivariate Gaussian Mixture Models (GMM) to the empirical $\log_2$ fold changes independently of the ± 0.5 cutoffs.

The input is a matrix $X \in \mathbb{R}^{N\times 2}$ ($N = 943$ core matched pairs), where for each gene i:

$$X_i = \begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{pmatrix} \log_2 \mathrm{FC}_{\mathrm{RNA},i} \\ \log_2 \mathrm{FC}_{\mathrm{Prot},i} \end{pmatrix}. \tag{1}$$

2. Model Specification and Biophysical Constraints

The probability density of observation X_i under a mixture of $K \in \{1, 2, 3\}$ bivariate Gaussians is:

$$p(X_i \mid \theta) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}_2 \left(X_i \mid \mu_k, \, \Sigma_k\right), \quad \text{with } \pi_k \geq 0, \quad \sum_{k=1}^{K} \pi_k = 1, \tag{2}$$

where $\mu_k = (\mu_{xk}, \mu_{yk})^\top$ are component centroids and $\Sigma_k = \begin{pmatrix} \sigma_{xk}^2 & \rho_k \sigma_{xk} \sigma_{yk} \\ \rho_k \sigma_{xk} \sigma_{yk} & \sigma_{yk}^2 \end{pmatrix}$ are covariance matrices.

To prevent unconstrained components from fitting arbitrary noise, three biophysical constraints were enforced:

1. **Component 1 (Diagonal / Co-regulation):** Represents canonical stoichiometric coupling. The correlation is bounded within a biologically realistic coupling window:

$$\rho_1 \in [0.70, \, 0.90]. \tag{3}$$

2. **Component 2 (Vertical / Protein-dominant decoupling):** Represents protein changes occurring without significant transcriptomic shifts. Coordinate correlation is set to zero, and variance is vertically elongated by factor c:

$$\rho_2 = 0, \qquad \sigma_{y2}^2 \geq c \cdot \sigma_{x2}^2. \tag{4}$$

3. **Component 3 (Horizontal / Post-transcriptional buffering, for $K = 3$):** Represents large mRNA changes buffered at the protein level. Correlation is set to zero, and variance is horizontally elongated:

$$\rho_3 = 0, \qquad \sigma_{x3}^2 \geq c \cdot \sigma_{y3}^2. \tag{5}$$

The elongation factor was systematically tested across $c \in \{1.3, 1.5, 2.0, 2.5, 4.0\}$ (aspect ratios $\sqrt{c} \in [1.14, 2.00]$).

3. Expectation-Maximization (EM) Algorithm

Parameters were estimated iteratively via the EM algorithm until the log-likelihood converged ($|\ell^{(t+1)} - \ell^{(t)}| < 10^{-6}$):

- **Initialization:** Posterior weights $\gamma_{ik}^{(0)} \in [0, 1]$ were initialized deterministically based on coarse regions (vertical candidates: $|x_i| < 0.55, |y_i| > 0.40$; horizontal candidates: $|x_i| > 0.50, |y_i| < 0.50$; diagonal otherwise), normalized so $\sum_k \gamma_{ik}^{(0)} = 1$.
- **M-Step (Maximization):** Given weights γ_{ik}, component sample sizes $N_k = \sum_i \gamma_{ik}$ and mixing weights $\pi_k = N_k/N$ were computed. Centroids and sample dispersions were updated:

$$\mu_k = \frac{1}{N_k} \sum_{i=1}^{N} \gamma_{ik} X_i, \quad v_{xk} = \frac{1}{N_k} \sum_{i=1}^{N} \gamma_{ik} (x_i - \mu_{xk})^2, \quad v_{yk} = \frac{1}{N_k} \sum_{i=1}^{N} \gamma_{ik} (y_i - \mu_{yk})^2. \tag{6}$$

  Constraints were then directly applied:

  - For $k = 1$: $\rho_1 = \min(0.90, \max(0.70, c_{xy,1}/\sqrt{v_{x1} v_{y1}}))$, and Σ_1 is assembled.
  - For $k = 2$: $\Sigma_2 = \mathrm{diag}(v_{x2},\, \max(v_{y2},\, c \cdot v_{x2}))$.
  - For $k = 3$: $\Sigma_3 = \mathrm{diag}(\max(v_{x3},\, c \cdot v_{y3}),\, v_{y3})$.

- **E-Step (Expectation):** Using the updated parameters, posterior responsibilities were recalculated:

$$\gamma_{ik}^{(t+1)} = \frac{\pi_k \, \mathcal{N}_2 \left(X_i \mid \mu_k,\, \Sigma_k\right)}{\sum_{j=1}^{K} \pi_j \, \mathcal{N}_2 \left(X_i \mid \mu_j,\, \Sigma_j\right)}, \quad \ell^{(t+1)} = \sum_{i=1}^{N} \ln \left(\sum_{k=1}^{K} \pi_k \, \mathcal{N}_2 \left(X_i \mid \mu_k,\, \Sigma_k\right) \right). \tag{7}$$

4. Model Selection (BIC)

Model selection across $K \in \{1, 2, 3\}$ and $c \in \{1.3, 1.5, 2.0, 2.5, 4.0\}$ was evaluated using the Bayesian Information Criterion:

$$\mathrm{BIC} = -2\hat{\ell} + P \ln(N), \tag{8}$$

where $\hat{\ell}$ is the maximized log-likelihood, $N = 943$ ($\ln(943) \approx 6.849$), and P is the number of free parameters:

- $K = 1$: 2 means + 2 variances + 1 covariance $\implies P = 5$.
- $K = 2$ (constrained): 1 weight (π_1) + 4 means + 3 dispersion (Comp 1) + 2 variances (Comp 2) $\implies P = 10$.
- $K = 3$ (constrained): 2 weights + 6 means + 3 (Comp 1) + 2 (Comp 2) + 2 (Comp 3) $\implies P = 15$.

BIC was used instead of classical likelihood-ratio tests because testing component complexity (K vs. $K + 1$) places the null hypothesis ($\pi_{K+1} = 0$) on the boundary of the probability simplex, violating Wilks' theorem regularity conditions.

5. Confidence Ellipses

Concentration ellipses for component k at confidence levels $\alpha \in \{0.50, 0.90, 0.95\}$ were plotted using the spectral decomposition $\hat{\Sigma}_k = V \Lambda V^\top$:

$$E_k(\alpha) = \hat{\mu}_k + \sqrt{F_{\chi^2_2}^{-1}(\alpha)} \cdot V \Lambda^{1/2} \begin{pmatrix} \cos\phi \\ \sin\phi \end{pmatrix}, \quad \phi \in [0, 2\pi), \tag{9}$$

with radii $\sqrt{F_{\chi^2_2}^{-1}(0.50)} \approx 1.177$, $\sqrt{F_{\chi^2_2}^{-1}(0.90)} \approx 2.146$, and $\sqrt{F_{\chi^2_2}^{-1}(0.95)} \approx 2.448$.

The complete analysis pipeline is implemented in base R and `data.table` in `Generate_Supplementary_Figure`